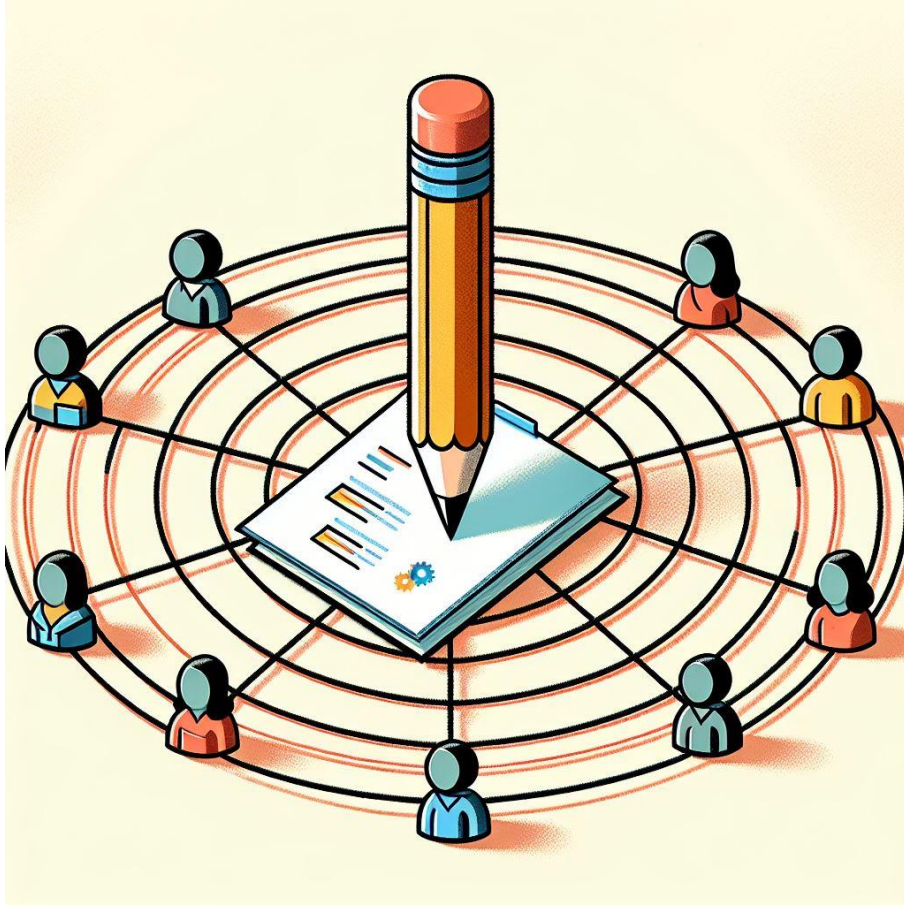


Dynamic Report Creation with RMarkdown

June 2025
Lindsay Dickey

CORE





CORE



Evidence *for* Change

Task: generate customized neighborhood reports from similar / same data

- Include neighborhood logos
- data results and charts
- comparisons to overall
- Include standard methodology and intro text
- Funder's branding , style
- Can be printed or emailed and read by all neighborhood organizations



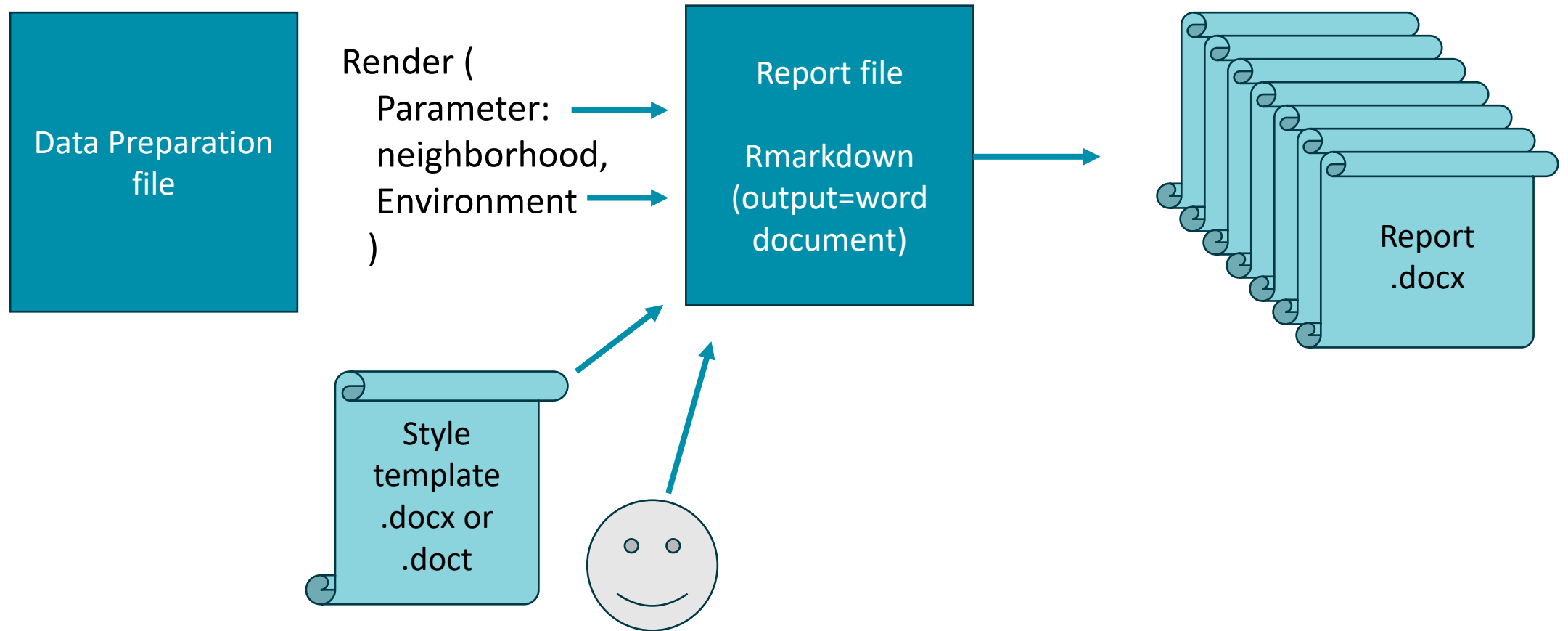
Solution: Parameterized reporting with RMarkdown

```
graph LR; A[Data Preparation] --> B["Report file (.rmd)  
output=word document"]
```

Data
Preparation

Report file
(.rmd)
output=word
document

Solution: Parameterized reporting with RMarkdown



Render function

```
#Single Test Case (recommended as you draft your repo
rmarkdown::render(
  input = "Neighborhood_Data_Report.rmd",
  params = list(neighborhood_var = "Greenwood"),
  envir = parent.frame(),
  output_file = paste0("Report_Greenwood.docx")
)

# Render for all units
for (unit in unique(metadata$neighborhood)) {
  rmarkdown::render(
    input = "Neighborhood_Data_Report.rmd",
    params = list(neighborhood_var = unit),
    envir = parent.frame(),
    output_file = paste0("Report_", unit, ".docx")
  )
}
```

Report Code- Yaml

```
---  
title: "The State Greenspace Foundation"  
subtitle: "Report for `r params$neighborhood_var`"  
date: "`r format(Sys.Date(), '%B %d, %Y')`"  
params:  
  neighborhood_var: "Greenwood"  
output:  
  word_document:  
    reference_docx: "StyleTemplate.docx"  
---
```

The State Greenspace Foundation

Report for Greenwood

June 17, 2025



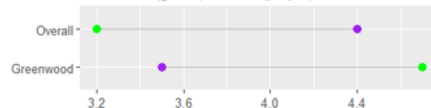
What we did

The State Greenspace Foundation funded community partners, including **Greenwood**, to radically improve the beauty of our region. Your neighborhood committee spent 100,000 to make your communities more accessible and attractive. **Greenwood** spent money on **Maintenance**. In 2019, when your neighborhood learned of their selection for the program, a neighborhood committee formed and surveyed residents of each neighborhood through posted QR codes. Three years after money was distributed, the committee engaged residents in a similar survey. This report highlights the evolving perspectives of your neighbors regarding your greenspace and compares the impact to the program impact across all funded communities in the state. Of the population of **Greenwood** (5000 in 2023), **217** people took the survey, or **4.3 %** of the population.

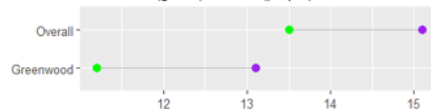
2

What we found

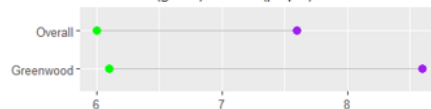
Days a Week Spent in Greenspace
from 2020 (green) to 2023 (purple)



Years in Your Home
from 2020 (green) to 2023 (purple)



Satisfaction with Neighborhood (1-10)
from 2020 (green) to 2023 (purple)



Resources

Want to learn more about community green space improvement in **Greenwood** ? Contact John Muir at 123-456-7890 or visit our website for more information.

The State Greenspace Foundation

Report for Farmville

June 17, 2025



The State Greenspace Foundation

Report for Beachtown

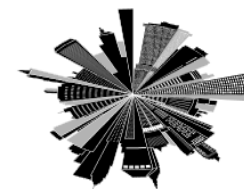
June 17, 2025



The State Greenspace Foundation

Report for Centerville

June 17, 2025



What we did

The State Greenspace Foundation funded community partners, including **Centerville**, to radically improve the beauty of our region. Your neighborhood committee spent 100,000 to make your communities more accessible and attractive. **Centerville** spent money on **Seedlings**. In 2019, when your neighborhood learned of their selection for the program, a neighborhood committee formed and surveyed residents of each neighborhood through posted QR codes. Three years after money was distributed, the committee engaged residents in a similar survey. This report highlights the evolving perspectives of your neighbors regarding your greenspace and compares the impact to the program impact across all funded communities in the state. Of the population of **Centerville** (10^4) in 2023, **198** people took the survey, or **2 %** of the

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mmittee spent 100,000 to
s spent money on **Shade**. In
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rough posted QR codes.
esidents in a similar survey.

Some References for Making R Markdown Reports

[Tips and tricks for working with images and figures in R Markdown documents | Technical Tidbits From Spatial Analysis & Data Science](#)

[render markdown with paths relative to output directory · Issue #587 · rstudio/rmarkdown](#)

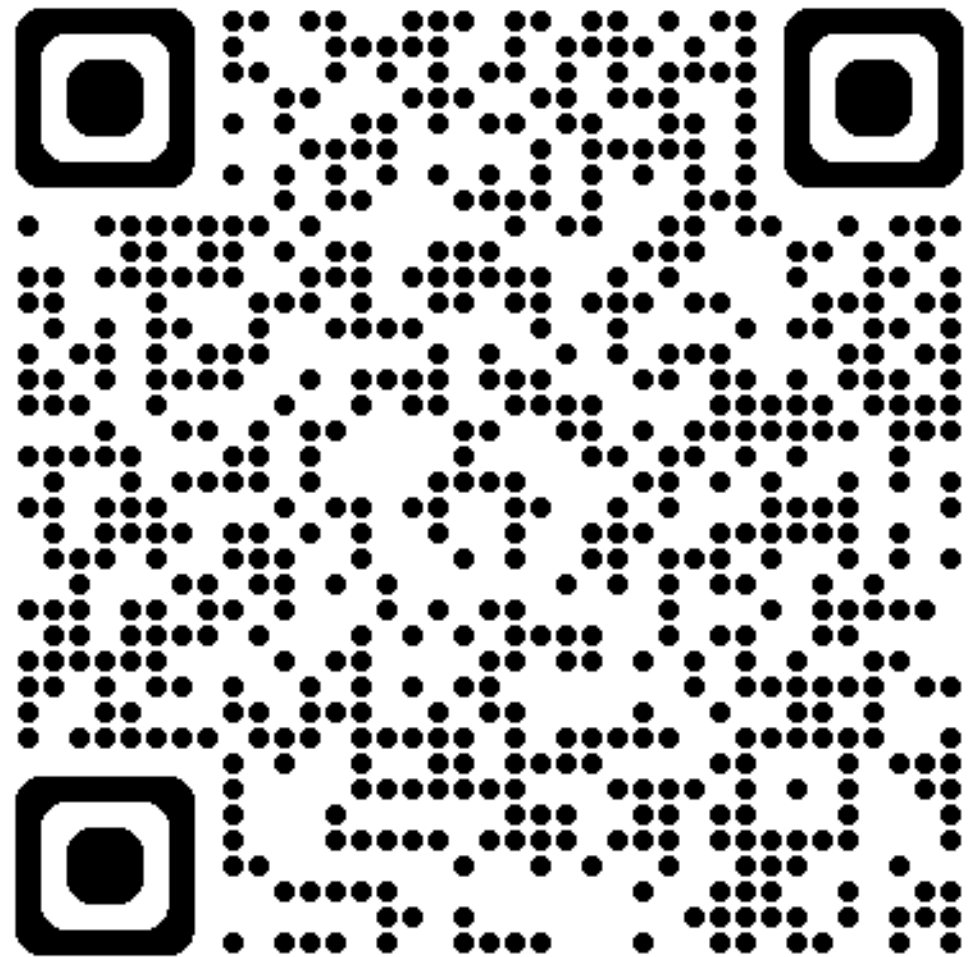
[How to use Quarto for Parameterized Reporting – Mike Mahoney](#)

[Simple way to Install Tex \(LATEX\) to export Rmarkdown to PDF \(in RStudio\) - YouTube](#)

Neighborhood
Report
Demonstration
Code on Megan's
github repository

Thank you, Megan!

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